

MICROCONTROLLER-BASED SOLAR CHARGER



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As the sources of conventional energy deplete day by day, resorting to alternative sources of energy like solar and wind energy has become need of the hour.

Solar-powered lighting systems are already available in rural as well as urban areas. These include solar lanterns, solar home lighting systems, solar streetlights, solar garden lights and solar power packs. All of them consist of four components: solar photovoltaic module, rechargeable battery, solar charge controller and load.

In the solar-powered lighting system, the solar charge controller plays an important role as the system's

overall success depends mainly on it. It is considered as an indispensable link between the solar panel, battery and load.

The microcontroller-based solar charge controller described here has the following features:

1. Automatic dusk-to-dawn operation of the load
2. Built-in digital voltmeter (0V-20V range)
3. Parallel- or shunt-type regulation
4. Overcharge protection
5. System status display on LCD
6. Deep-discharge protection
7. Low battery lock
8. Charging current changes to 'pulsed' at full charge

9. Low current consumption
10. Highly efficient design based on microcontroller

11. Suitable for 10-40W solar panels for 10A load

The circuit of the solar charge controller is shown in Fig. 1. It comprises microcontroller AT89C2051, serial analogue-to-digital converter ADC0831, optocoupler MCT2E, regulator 7805, MOSFETs

BS170 and IRF540N, transistor BC547, LCD and a few discrete components. Component description is

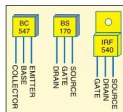


Fig. 2: Pin configurations of BC547, BS170 and IRF540

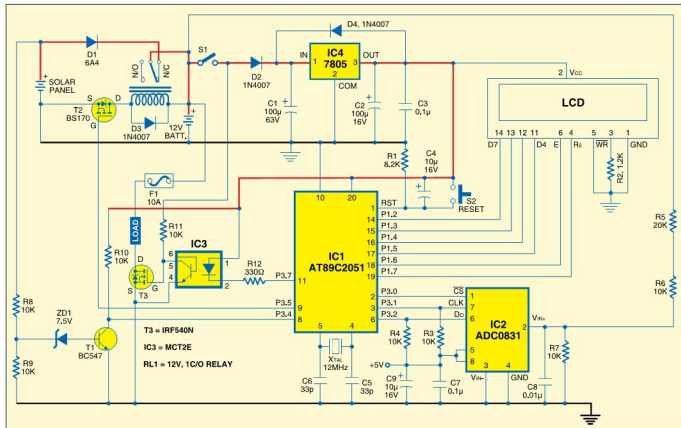


Fig. 1: Circuit of microcontroller-based solar charger